2016.0 RANGE ROVER (LG), 412-01

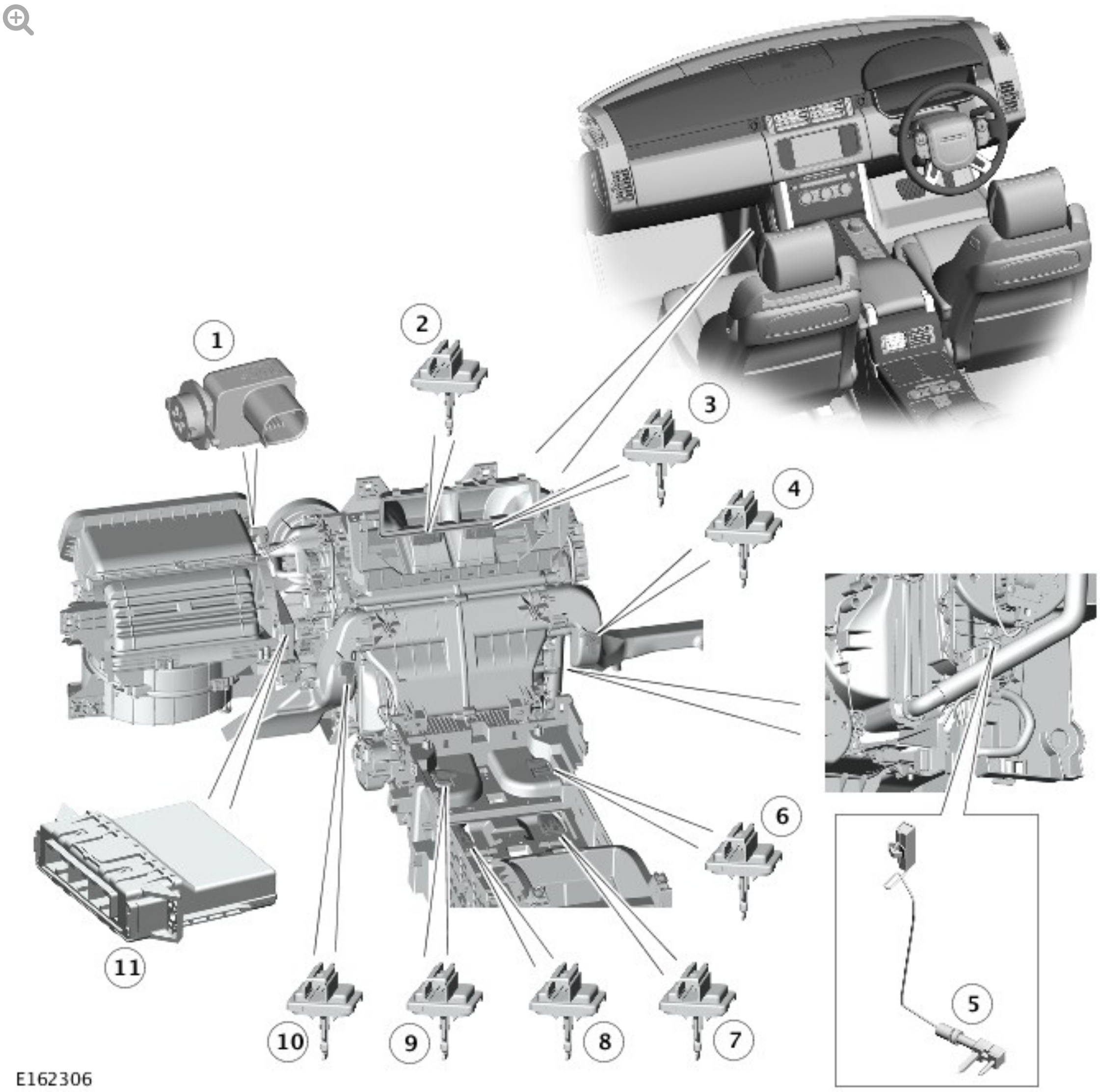
CLIMATE CONTROL

DESCRIPTION AND OPERATION

COMPONENT LOCATION - SHEET 1 OF 3

NOTE:

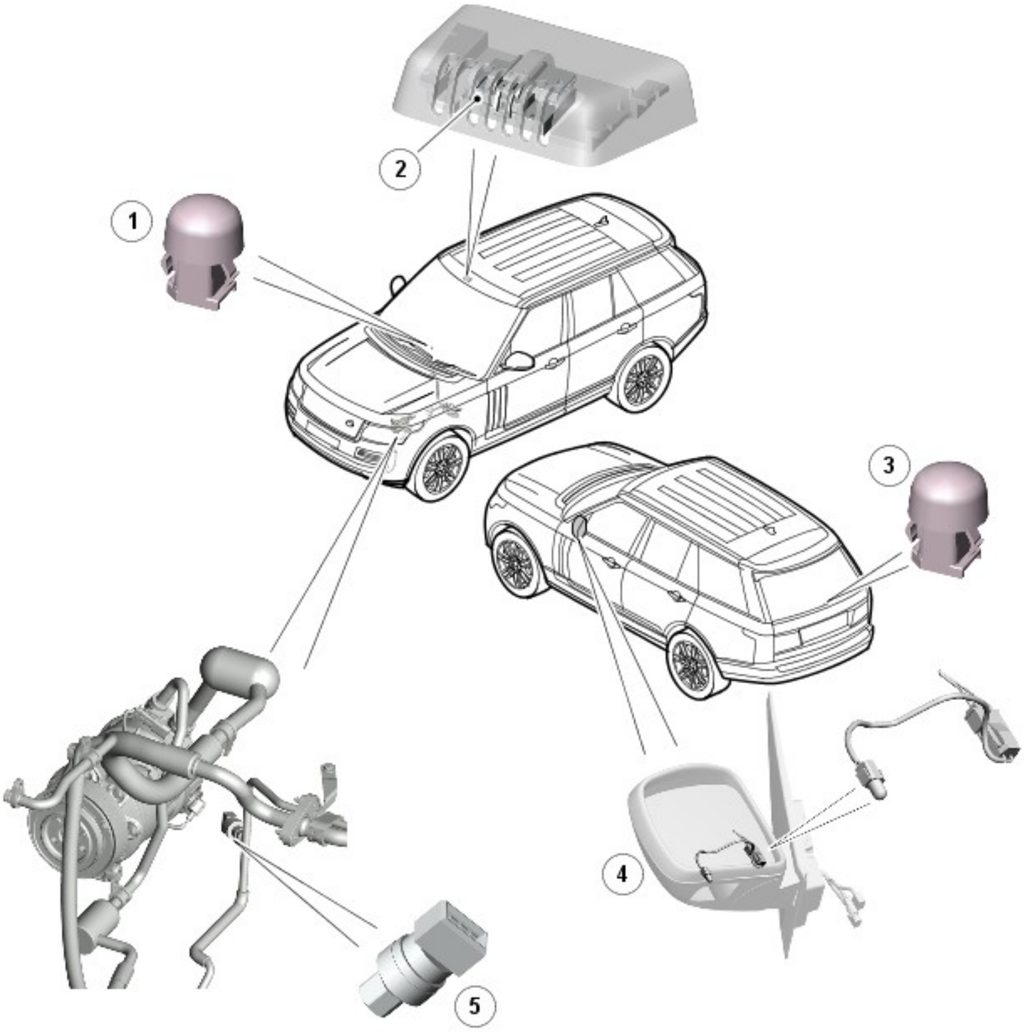
RHD (right-hand drive) installation shown, LHD (left-hand drive) installation similar.



ITEM	DESCRIPTION
1	Pollution sensor
2	Duct air temperature sensor - left front face
3	Duct air temperature sensor - right front face
4	Duct air temperature sensor - right front foot
5	Evaporator temperature sensor
6	Duct air temperature sensor - right rear foot
7	Duct air temperature sensor - right rear face
8	Duct air temperature sensor - left rear face
9	Duct air temperature sensor - left rear foot
10	Duct air temperature sensor - left front foot

11	Automatic temperature control module

COMPONENT LOCATION – SHEET 2 OF 3



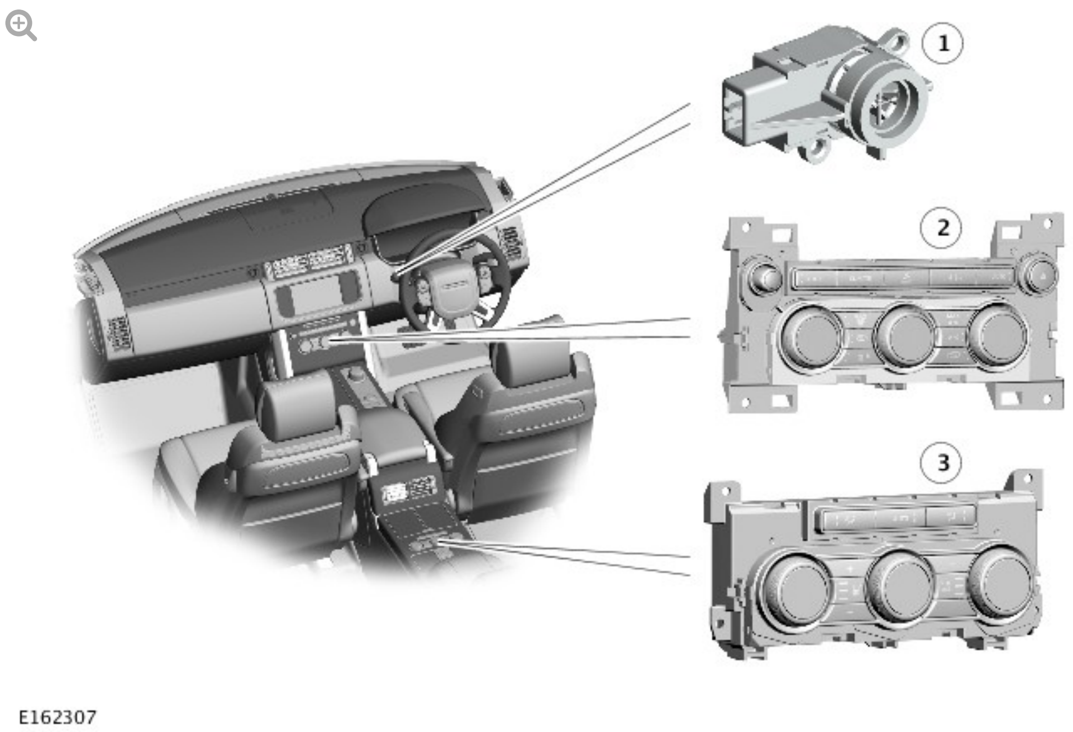
E141124

ITEM	DESCRIPTION
1	Front sunload sensor
2	Humidity sensor
3	Rear sunload sensor
4	Ambient air temperature sensor
5	Refrigerant pressure sensor (TDV6 3.0L diesel installation shown, other installations similar)

COMPONENT LOCATION - SHEET 3 OF 3

NOTE:

RHD installation shown, LHD installation similar.



ITEM	DESCRIPTION
1	In-vehicle temperature sensor
2	Integrated Control Panel (ICP)
3	Rear ICP

OVERVIEW

The standard climate control system operates the A/C (air conditioning) system and the heating and ventilation system to control the temperature, volume and distribution of air from the climate control assembly. Operation can be fully automatic, but manual overrides are provided for the intake air source, blower speed and air distribution.

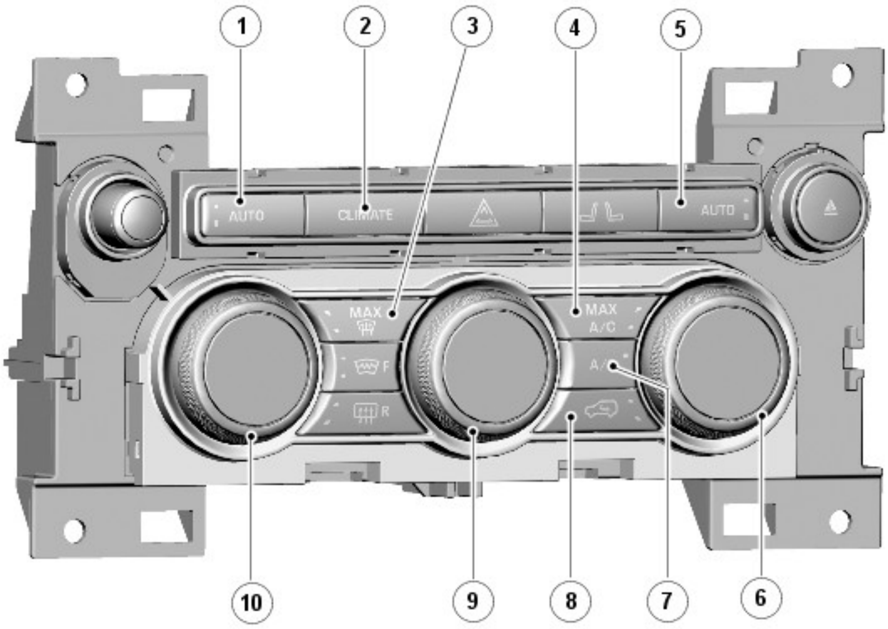
Auxiliary climate control provides individual control for the left and right rear seat passenger areas to give a four zone climate control system. For additional information, refer to: Auxiliary Climate Control (412-02A Auxiliary Climate Control, Description and Operation).

The climate control system consists of the following components:

- Integrated Control Panel (ICP)
- Rear ICP
- Automatic Temperature Control Module (ATCM)
- Ambient Air Temperature (AAT) sensor
- Refrigerant pressure sensor
- Evaporator temperature sensor
- In-vehicle temperature sensor
- Humidity sensor
- Front and rear sunload sensors
- Pollution sensor
- Duct air temperature sensors.

DESCRIPTION

INTEGRATED CONTROL PANEL (ICP)



E141125

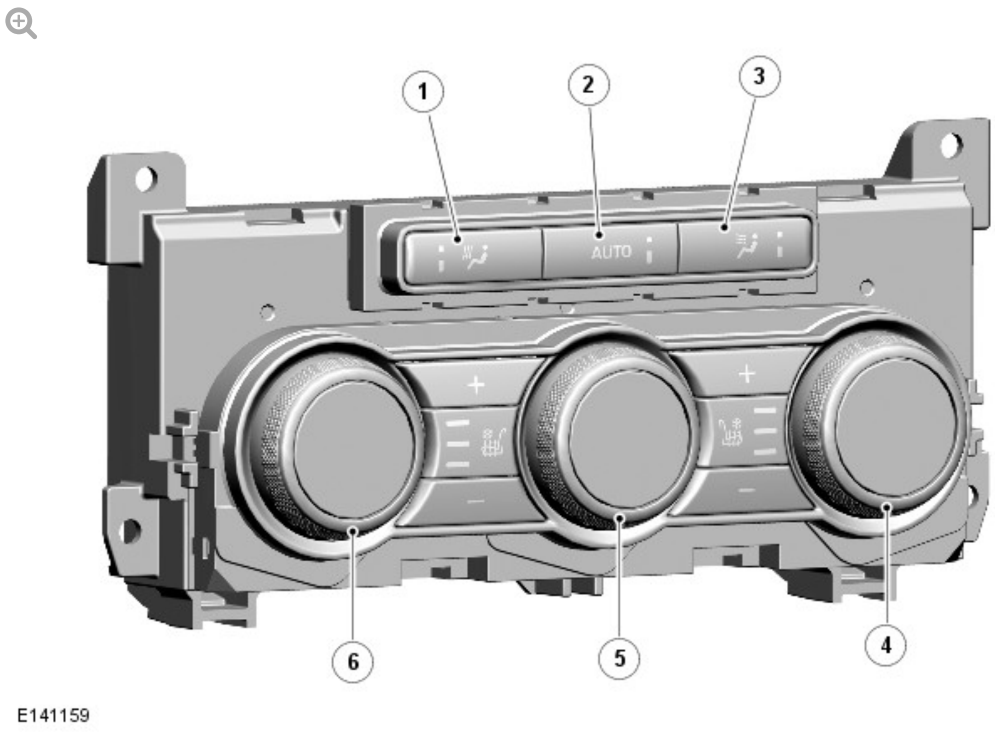
ITEM	DESCRIPTION
1	Left automatic mode switch
2	Climate control menu switch
3	Maximum demist switch
4	Maximum air conditioning switch
5	Right automatic mode switch
6	Right temperature switch
7	Air conditioning system on/off switch
8	Recirculation switch
9	Blower switch
10	Left temperature switch

The Integrated Control Panel (ICP) is installed in the center of the instrument panel and contains push switches and rotary switches for operation of the climate control system. Each push switch contains an amber status LED (light emitting diode). Display screens are integrated into the center of the rotary switches to display the selected temperature and blower speed as applicable. In addition to the switches for the climate control system, the ICP also contains switches for operation of:

- The audio system.
  For additional information, refer to: Audio System (415-01 Information and Entertainment System, Description and Operation).
- The hazard warning indicators.
  For additional information, refer to: Exterior Lighting (417-01 Exterior Lighting, Description and Operation).
- The front seat heaters/climate system.
  For additional information, refer to: Seats (501-10 Seating - Short Wheelbase, Description and Operation).
- The windshield and rear window heaters.
  For additional information, refer to: Glass, Frames and Mechanisms (501-11 Glass, Frames and Mechanisms, Description and Operation).

The ICP converts selections on the climate control switches into medium speed CAN (controller area network) messages and transmits them to the ATC (automatic temperature control) module.

REAR INTEGRATED CONTROL PANEL



ITEM	DESCRIPTION
1	Foot distribution switch
2	Automatic mode switch
3	Face distribution switch
4	Right temperature switch
5	Blower switch
6	Left temperature switch

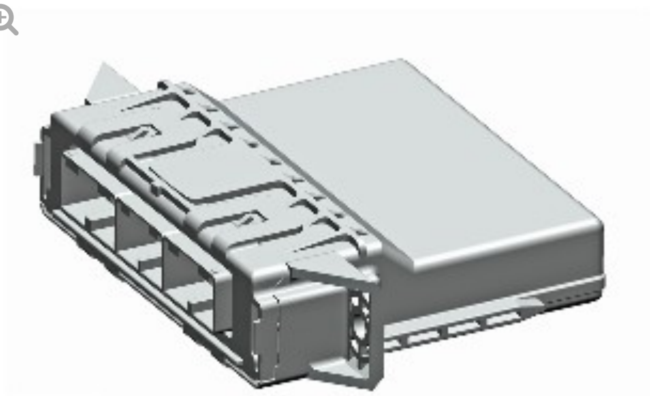
On the Autobiography Black vehicle the rear ICP is installed in the rear floor console. The ICP contains rotary and push button switches for controlling the temperature, volume and distribution of air to the rear seats. Each push switch contains an amber status LED. Display screens are integrated into the center of the rotary switches to display the selected temperature and blower speed as applicable.

Additional push switches on the rear ICP are provided for control of seat heating/climate control seats.
For additional information, refer to: Seats (501-10 Seating - Short

Wheelbase, Description and Operation).

The rear ICP has a permanent battery power feed from the CJB (central junction box), and is connected to the medium speed CAN bus. The rear ICP converts selections on the climate control switches into medium speed CAN messages and transmits them to the ATC module.

AUTOMATIC TEMPERATURE CONTROL MODULE



E141126

The ATC module is mounted on the inboard side of the air inlet duct, behind the front passenger side of the instrument panel.

The ATC module processes inputs from the climate control system sensors, the ICP, the rear ICP, the touch screen and other systems, then outputs the appropriate control signals to the A/C system and the heating and ventilation system. In addition to the A/C system and the heating and ventilation system, the ATC module also controls the following:

- The windshield washer jet heaters and the wiper blade heater.
  For additional information, refer to: Wipers and Washers (501-16 Wipers and Washers, Description and Operation).
- The front and rear seat heaters (where fitted).
  For additional information, refer to: Seats (501-10 Seating - Short Wheelbase, Description and Operation).
- Auxiliary climate control system (where fitted).
  For additional information, refer to: Auxiliary Climate Control (412-02A Auxiliary Climate Control, Description and Operation).

- Fuel fired booster heater (where fitted).
  For additional information, refer to: Fuel Fired Booster Heater (412-02 Auxiliary Climate Control - GTDi 2.0L Petrol, Description and Operation).
- Electric booster heater (where fitted).
  For additional information, refer to: Auxiliary Climate Control (412-02A Auxiliary Climate Control, Description and Operation).

AMBIENT AIR TEMPERATURE SENSOR

The ambient air temperature sensor is a NTC (negative temperature coefficient) thermistor installed in the left exterior rear view mirror, with the bulb of the sensor positioned over a hole in the bottom of the mirror casing.

The ECM (engine control module) supplies the ambient air temperature sensor with a 5 V reference voltage and translates the return signal voltage into a temperature. The ECM transmits the temperature on the high speed CAN bus for use by other systems. The ATC module receives the temperature, on the medium speed CAN bus, from the gateway module.

REFRIGERANT PRESSURE SENSOR

The refrigerant pressure sensor is located in the high pressure refrigerant line between the condenser and the thermostatic expansion valve.

The ATC module supplies a 5 V reference voltage to the refrigerant pressure sensor and receives a return signal voltage related to system pressure.

The ATC module uses the signal from the refrigerant pressure sensor to protect the refrigerant system from extremes of pressure and to calculate the A/C compressor load on the engine. The ATC module transmits the A/C compressor load value to the ECM, via the medium speed CAN bus, gateway module and high speed CAN bus, for use in controlling the speed of the engine cooling fan.

To protect the system from extremes of pressure, the ATC module sets the A/C compressor to the minimum flow position if the pressure:

- Decreases to 1.9 ± 0.2 bar (27.5 ± 3 lbf/in²); the ATC module loads the A/C compressor again when the pressure increases to 2.8 ± 0.2 bar (40.5 ± 3 lbf/in²).
- Increases to 33 ± 1 bar (479 ± 14.5 lbf/in²); the ATC module loads the A/C compressor again when the pressure decreases to 23.5 ± 1 bar (341 ± 14.5 lbf/in²).

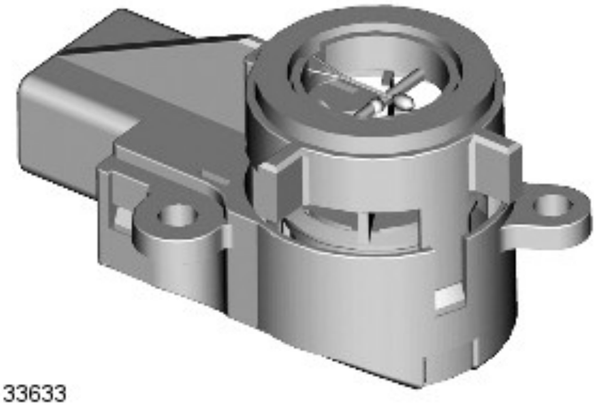
EVAPORATOR TEMPERATURE SENSOR

The evaporator temperature sensor is a NTC thermistor that provides the ATC module with a temperature signal from the air outlet side of the evaporator. The sensor extends into the air flow on the downstream side of the evaporator.

The ATC module uses the input from the evaporator temperature sensor to control the load of the A/C compressor, and thus the operating temperature of the evaporator.

The ATC module supplies the evaporator temperature sensor with a 5 V reference voltage and translates the return signal voltage into a temperature. If the sensor develops a fault, the ATC module adopts a default temperature of 0 °C (32 °F).

IN-VEHICLE TEMPERATURE SENSOR



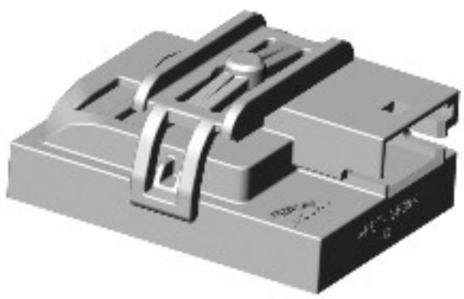
E133633

The in-vehicle temperature sensor is a NTC thermistor installed behind a grill in the instrument panel, on the inboard side of the steering column. A motor driven fan in the sensor draws air through the grill and over the thermistor.

The ATC module uses the signal from the in-vehicle temperature sensor for control of the climate control assembly output temperatures, blower speed and air distribution.

The ATC module supplies the in-vehicle temperature sensor with a 5 V reference voltage and translates the return signal voltage into a temperature. If the in-vehicle temperature sensor develops a fault, the ATC module adopts a default temperature of 20 °C (68 °F).

HUMIDITY SENSOR



E141130

The humidity sensor is installed in a bracket attached to the inside of the windshield, to the right of the interior mirror. The sensor is concealed under a cover, which clips onto the bracket. The sensor comprises three individual elements:

- A capacitive humidity sensor
- A NTC thermistor air temperature sensor
- An infrared windshield glass temperature sensor.

The humidity sensor is powered by a feed from the battery saver relay in the CJB. The data from the three individual elements of the humidity sensor are transmitted in LIN (local interconnect network) bus messages to the ATC module. From the data, the ATC module:

- Adjusts the humidity of the air in the vehicle as necessary, to provide the optimum comfort level for occupants
- Calculates the dew point temperature of the air at the inside of the windshield.

Humidity within the vehicle is controlled by raising or lowering the temperature of the evaporator. An increase in evaporator temperature increases the moisture content of the air at the windshield. Lowering the evaporator temperature reduces the moisture content of the air in the interior.

If the temperature of the windshield glass is equal to or less than the dew point temperature, misting is likely to occur. To prevent this, the ATC module will:

- Increase the temperature of the air leaving the climate control assembly
- Adjust the position of the demist motor to direct more air to the windshield
- Adjust the position of the recirculation motor to admit more fresh air
- Energize the windshield heater.

SUNLOAD SENSORS

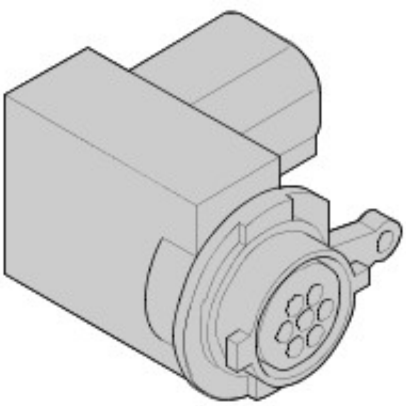
E141131

Front and rear sunload sensors are installed. The front sunload sensor is installed in the top of the instrument panel, in the center demist grille. The rear sunload sensor is installed in the interior trim panel of the upper tailgate.

The ATC module supplies the sunload sensors with a 5 V reference voltage. Each sensor contains photoelectric cells that provide the ATC module with two inputs of light intensity, which equates to the solar heating effect on the left and right sides of the vehicle.

The ATC module compensates for the solar heating effect by adjusting blower speed, climate control assembly output temperature and air distribution to maintain the required zone temperatures.

POLLUTION SENSOR



E133637

The sensor is mounted on the inboard side of the air inlet duct, in the fresh air inlet.

The pollution sensor has battery power, ground and signal connections with the ATC module. Semiconductor material in the sensor undergoes a change of conductance if exposed to specific gases. The sensor uses this property to monitor for gases such as CO and NOx and so determine the quality of the fresh air entering the vehicle. The sensor transmits the air quality to the ATC module, in a PWM (pulse width modulation) signal, as one of the following four conditions:

- Static or reduced pollution levels
- Small increase in pollution levels
- Medium increase in pollution levels
- Rapid or large increase in pollution levels.

Based on the signal from the pollution sensor, the ATC module controls the intake air source to reduce the amount of contaminants entering the interior. This function is fully automatic, but can be overridden by manual selection of the air intake source using the fresh/re-circulated air switch on the ICP.

DUCT AIR TEMPERATURE SENSORS

E141132

The duct air temperature sensors are NTC thermistors installed in the following distribution ducts:

- Left and right center front face ducts
- Left and right front foot ducts

- Left and right second row foot ducts
- Left and right second row face ducts

The duct air temperature sensors are provided with 5V reference signal and ground connections with the ATC module. The ATC module uses the sensor inputs in the calculations used to set the temperatures of the air leaving the climate control assembly.

OPERATION

When the engine is running, the standard climate control system can be activated using the AUTO switches on the ICP, or the AUTO soft keys on the **Front climate** or **Rear climate** menus of the touch screen. Provided the rear climate control panel is unlocked (on the touch screen **Rear climate** menu), pressing the AUTO switch on the rear ICP will also activate the system.

INTAKE AIR CONTROL

The source of inlet air is automatically controlled by the ATC module, unless overridden by pressing the recirculation switch on the ICP to give timed or latched recirculation. A brief press of the switch illuminates the switch indicator and activates timed recirculation. Pressing and holding the switch causes the switch indicator to flash and then illuminate constantly, indicating that the air inlet is in latched recirculation and the switch can be released. A second press of the switch cancels recirculation and the ATC module returns the recirculation door to the fresh air position. Timed recirculation is automatically cancelled after a set time, which varies with ambient air temperature.

During automatic control, the ATC module determines the required position of the recirculation door from comfort algorithms and the pollution sensor. If it detects pollution, the ATC module sets the air source to recirculation for 10 minutes, then to fresh air for 20 seconds to renew the air in the vehicle. The ATC module repeats this cycle until the pollution is no longer present.

The sensitivity of the pollution sensor can be adjusted on the touch screen, using the **Settings** soft key of the **Front climate** menu. The pollution sensing function can also be switched off by adjusting sensitivity to the minimum setting. If there is a fault with the pollution sensor, the ATC module disables automatic operation of the recirculation door.

AIR TEMPERATURE CONTROL

The ATC module calculates the temperature blend motor positions required to achieve the selected temperature and compares it against the current position. If there is any difference, the ATC module signals the motors to adopt the new position.

Air temperature is controlled automatically unless maximum heating (HI) or maximum cooling (LO) is selected. When maximum heating or cooling is selected, a comfort algorithm in the ATC module adopts an appropriate strategy for air distribution, blower speed, and air source.

Temperature control in one zone can be compromised by another zone being set to a high level of heating or cooling. True maximum heating or cooling (displayed as **HI** or **LO** on the touch screen and switches) can only be selected for the driver zone. If HI or LO is selected for the driver zone, the temperature for the other zones is automatically set to match the driver zone.

If A/C is selected off in the automatic mode, no cooling of the inlet air will take place. The minimum output air temperature from the system will be ambient air temperature plus any heat pick up in the air inlet path.

If the **Sync** soft key on the touch screen **Front climate** menu is pressed, the ATC module synchronizes the temperature of the other zones with the driver zone temperature.

BLOWER CONTROL

When the system is in the automatic mode, the ATC module determines the blower speed required from a comfort algorithm. When the system is in the manual mode, the ATC module operates the blower at the speed selected on the ICP or the touch screen. The ATC module also adjusts blower speed to compensate for the ram effect on inlet air produced by forward movement of the vehicle. As vehicle speed and ram effect increases, blower motor speed is reduced, and vice versa.

The blower switch on the rear ICP has no control over the blower in the climate control assembly. Instead, the ATC module adjusts the rear distribution doors to increase the air flow through the rear distribution ducts from the climate control assembly.

AIR DISTRIBUTION CONTROL

When the A/C system is in automatic mode, the ATC module automatically controls air distribution into the passenger compartment in line with its comfort algorithm. Automatic control is overridden by distribution selections on the touch screen, the ICP or the rear ICP. Air distribution remains as selected until one of the AUTO is selected or a different manual selection is made.

AIR CONDITIONING COMPRESSOR CONTROL

When A/C is selected the ATC module maintains the evaporator at an operating temperature that varies with the passenger compartment cooling requirements. If the requirement for cooled air decreases, the ATC module raises the evaporator operating temperature by reducing the flow of refrigerant provided by the A/C compressor. The ATC module closely controls the rate of temperature increase to avoid introducing moisture into the passenger compartment.

If the requirement for cooled air increases, the ATC module lowers the evaporator operating temperature by increasing the flow of refrigerant provided by the A/C compressor.

When A/C is off, the compressor current signal supplied by the ATC module holds the A/C compressor control valve in the minimum flow position and disengages the clutch to switch off the A/C function.

The ATC module incorporates limits for the operating pressure of the refrigerant system. If the system approaches the high pressure limit, the compressor current signal is progressively reduced until the system pressure decreases. If the system falls below the low pressure limit, the compressor current signal is held at its lowest setting so that the A/C compressor is maintained at its minimum stroke. This avoids depletion of the lubricant from the A/C compressor.

AIR CONDITIONING COMPRESSOR TORQUE

The ATC module transmits refrigerant pressure and A/C compressor current values to the ECM on the medium speed CAN bus, then through the gateway module and onto the high speed CAN bus. The ECM uses these values to calculate the torque being used to drive the A/C compressor. The ECM compares the calculated value with its allowable value and, if necessary, forces the ATC module to inhibit the A/C compressor by transmitting the 'ACClutchInhibit' CAN message. This forces the ATC module to reduce the drive current to the A/C compressor control valve, which reduces refrigerant flow. This in turn reduces the torque required to drive the A/C compressor.

By reducing the maximum A/C compressor torque, the ECM is able to reduce the load on the engine when it needs to maintain vehicle performance or cooling system integrity.

COOLING FAN CONTROL

The ATC module determines the amount of condenser cooling required from the refrigerant pressure sensor, since there is a direct relationship between the temperature and pressure of the refrigerant. The cooling requirement is broadcast to the ECM on the medium speed CAN bus. The ECM then controls the temperature of the condenser using the cooling fan(s).

MAXIMUM DEMIST

When the MAX demist switch on the ICP is pressed, the ATC module configures the climate control system as follows:

- Automatic mode off.
- A/C on.

- Selected temperature unchanged.
- Air inlet set to fresh air.
- Air distribution set to windshield.
- Blower speed set to high speed.

The ATC module also sends a medium speed CAN message to the CJB to activate the windshield and rear window heaters.

The programmed demist function can be cancelled by one of the following:

- A second press of the MAX demist switch.
- Selecting any air distribution soft key on the touch screen.
- Pressing the driver side AUTO switch on the ICP.
- Switching the ignition OFF.

The blower speed can be adjusted without terminating the programmed demist function. If the blower speed has been adjusted and then the MAX demist switch is pressed again, the system will go back to the MAX demist default settings. If the MAX demist program is cancelled with the MAX demist switch or the driver side AUTO switch, the windshield and rear window heaters remain on until they time out.

REST HEAT (VEHICLES WITH FUEL FIRED BOOSTER HEATER ONLY)

When the engine is not running, pressing the **Rest heat** soft key on the touch screen **Front climate** menu activates the rest heat function to heat the passenger compartment with residual heat from the engine. The rest heat function activates provided that:

- It is less than 2 minutes since the ignition was selected off.
- Ambient air temperature is less than 15 °C (59 °F).
- On the previous ignition cycle engine temperature exceeded 70 °C (158 °F).
- Battery voltage is 11.4 V minimum.

To provide the rest heat function, the ATC module:

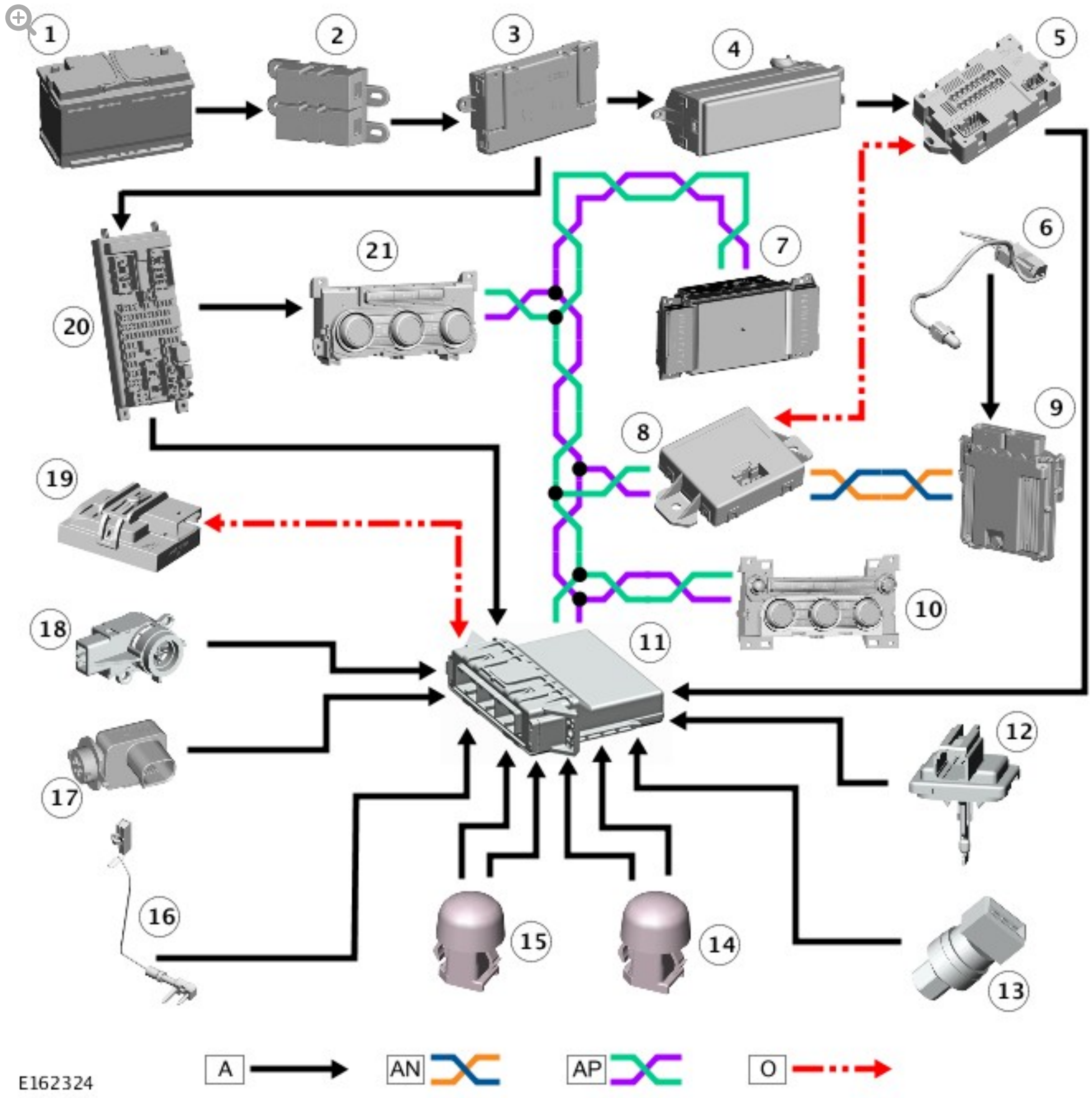
- Sends a LIN bus message to the FFBH (fuel fired booster heater) control module to run the FFBH coolant pump.
- Operates the blower at medium speed.
- Sets the distribution doors to direct air to the face outlets.
- Sets the temperature blend doors to produce the temperature selected on the driver side before the ignition was switched off.

The rest heat function is cancelled after 15 minutes or when:

- The **Rest heat** soft key is pressed again.
- The ignition is selected on.
- Battery voltage decreases to less than 11 V.

CONTROL DIAGRAM

Sheet 1 of 2

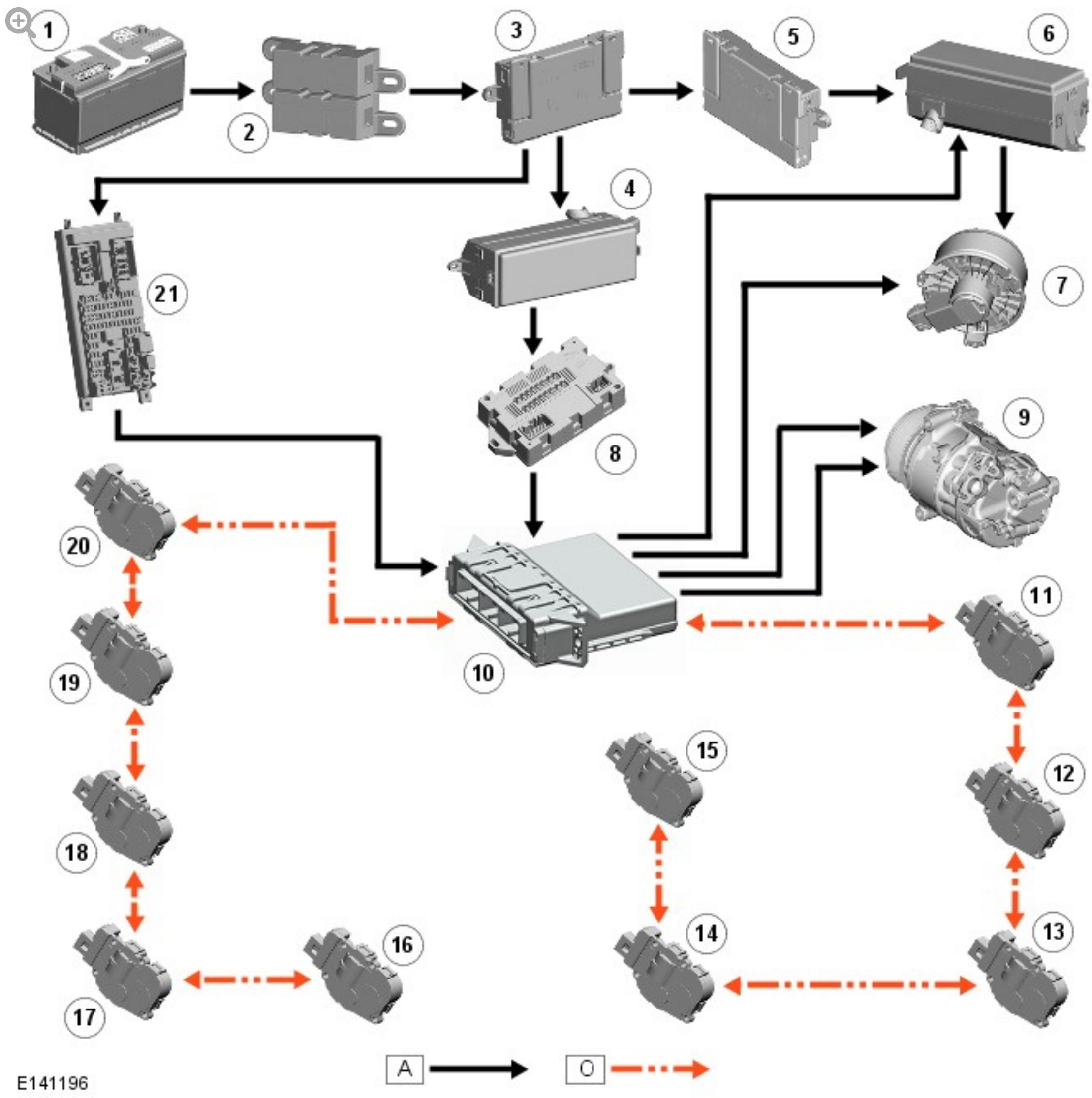


A = HARDWIRED; AN = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) POWERTRAIN BUS; MEDIUM SPEED (MS) CAN COMFORT BUS; O = LOCAL INTERCONNECT NETWORK (LIN) BUS.

ITEM	DESCRIPTION
1	Primary battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Quiescent Current Control Module (QCCM)
6	Ambient Air Temperature (AAT) sensor
7	Touch Screen (TS)

8	Gateway Module (GWM)
9	Engine Control Module (ECM)
10	Integrated Control Panel (ICP)
11	Automatic Temperature Control Module (ATCM)
12	Duct air temperature (8 off)
13	Refrigerant pressure sensor
14	Rear sunload sensor
15	Front sunload sensor
16	Evaporator temperature sensor
17	Pollution sensor
18	In-vehicle temperature sensor
19	Humidity sensor
20	Central Junction Box (CJB)
21	Rear Integrated Control Panel (ICP)

Sheet 2 of 2



A = HARDWIRED; O = LOCAL INTERCONNECT NETWORK (LIN) BUS.

ITEM	DESCRIPTION
1	Primary battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Auxiliary Junction Box (AJB)
6	Engine Junction Box (EJB)
7	Blower
8	Quiescent Current Control Module (QCCM)

9	Air Conditioning (A/C) compressor
10	Automatic Temperature Control Module (ATCM)
11	Recirculation motor
12	Cool air bypass motor
13	Distribution motor - right front face and feet
14	Temperature blend motor - right front
15	Temperature blend motor - right rear
16	Distribution motor - left front face and feet
17	Temperature blend motor - left front
18	Temperature blend motor - left rear
19	Distribution motor - rear face and feet
20	Distribution motor - demist
21	Central Junction Box (CJB)